

Agenda

Review of CIs

Prediction and tolerance intervals

Why CI?

A random variable X distributed according to a pdf with an unknown parameter

Interest: To identify the unknown parameter

CI gives information on the unknown parameter of the distribution by using a random sample

Main idea:

Collect a random sample $\{X_1, \dots, X_n\}$

Determine the parameter of interest, say θ

- Find a random variable, say W , that involves the random sample as well as the parameter θ and whose distribution you can specify, $W=h(X_1, \dots, X_n; \theta)$

Use the distribution of W to find random lower and upper bounds on θ

100(1- α) CIs around the true mean, μ

- $X_1, \dots, X_n$: A random sample with $E[X] = \mu$
- We are looking for a random variable $Y=h(X_1, \dots, X_n; \mu)$ whose distribution can be specified.

	X is normal		X is not normal		
	$n < 40$	$n \geq 40$	$n < 30$	$30 \leq n < 40$	$n \geq 40$
σ : known	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	?	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
σ : unknown	$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}$	$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$	?	?	$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$

$$\sigma = \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}$$

$$s = \sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}$$

100(1- α) CIs around $\mu_X - \mu_Y$

- $\{X_1, \dots, X_n\}, \{Y_1, \dots, Y_m\}$: Two random samples independent of each other with $E[X] = \mu_X$ and $E[Y] = \mu_Y$ with $\text{Var}(X) = \sigma_X$ and $\text{Var}(Y) = \sigma_Y$
- We are looking for a random variable $U = h(X_1, \dots, X_n; Y_1, \dots, Y_m; \mu_X, \mu_Y)$ whose distribution can be specified.

	X, Y : normal		X, Y: not normal		
	n,m<40	n,m $\geq$ 40	n,m<30	30 $\leq$ n,m<40	n,m $\geq$ 40
σ : known	$\bar{x} - \bar{y} \pm z_{\alpha/2} \sigma$	$\bar{x} - \bar{y} \pm z_{\alpha/2} \sigma$	?	$\bar{x} - \bar{y} \pm z_{\alpha/2} \sigma$	$\bar{x} - \bar{y} \pm z_{\alpha/2} \sigma$
σ : unknown	$\bar{x} - \bar{y} \pm t_{v,\alpha/2} s$	$\bar{x} - \bar{y} \pm z_{\alpha/2} s$	?	?	$\bar{x} - \bar{y} \pm z_{\alpha/2} s$

When to use student t statistics?

Check normality

- Look at the plots.
- Compute skewness.
- Do not use student t if there is an outlier.

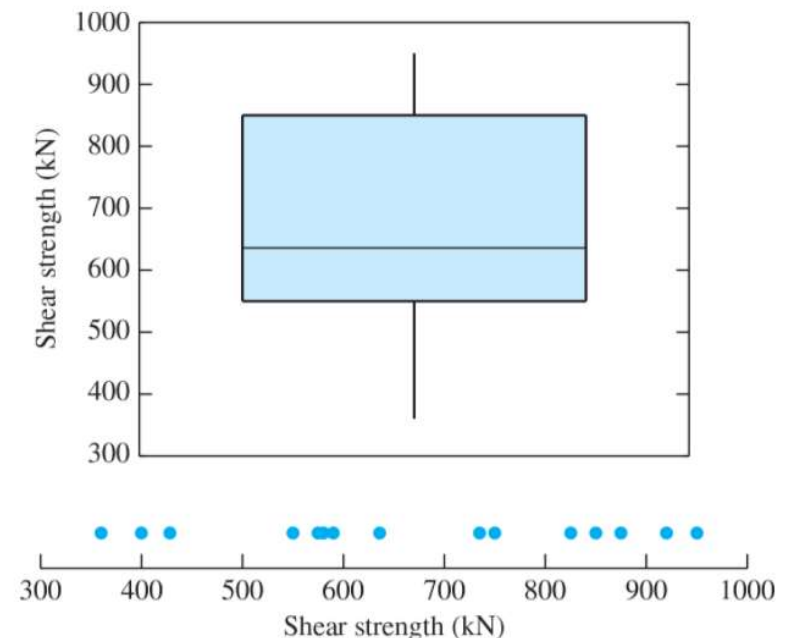
Example

The article “Direct Strut-and-Tie Model for Prestressed Deep Beams” presents measurements of the nominal shear strength (in kN) for a sample of 15 prestressed concrete beams. The results are

580	400	428	825	850	875	920	550
575	750	636	360	590	735	950	

Can you find a 95% CI around the mean?

Sample mean=668.27, $s=192.089$
520.62, 815.92



Example

An engineer reads a report that states that a sample of 11 concrete beams has an average compressive strength of 38.45 MPa with standard deviation 0.14 MPa. Should the t curve be used to find a confidence interval for the mean compressive strength?

Can you find a 95% CI around the mean?

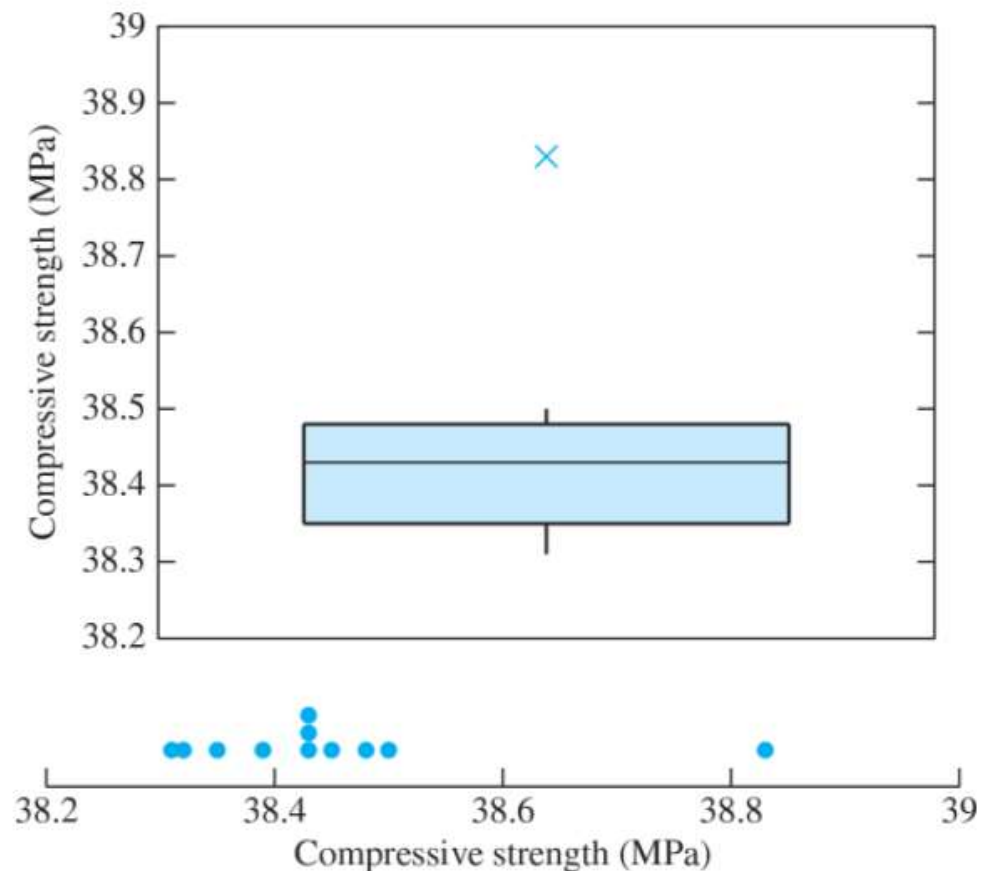
No info on the distribution, we cannot find.

Example

An engineer reads a report that states that a sample of 11 concrete beams has an average compressive strength of 38.45 MPa with standard deviation 0.14 MPa. Should the t curve be used to find a confidence interval for the mean compressive strength?

Can you find a 95% CI around the mean?

No because there is an outlier!



Correlated samples

	Samples with independence assumption (n=10)		Samples that take correlation into account (n=10)	
	Sample mean	Sample std deviation	Sample mean	Sample std deviation
X_i	12.77	4.17	12.77	4.17
Y_i	11.77	4.17	11.77	4.17
D_i	1	3.32	1	0

CIs? Conclusions?

100(1- α)CIs around the true variance of a random variable X

- $X_1, \dots, X_n$: A random sample with $\text{Var}(X) = \sigma^2$
- We are looking for a random variable $Y = h(X_1, \dots, X_n; \sigma)$ whose distribution can be specified.

X~Normal	X is not normal
All n	All n
$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2, n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2, n-1}} \right)$	

100(1- α) CIs around the true population proportion p

X : Number of successes in n trials, each trial is i.i.d. with constant $p = P\{\text{success}\}$

We are looking for a random variable $Y = h(X; \theta)$ whose distribution can be specified.

$$\hat{p} = \frac{X}{n}$$

$$\tilde{p} = \frac{X + 2}{n + 4}$$

$$\tilde{n} = n + 4$$

$$n\hat{p} \geq 10 \text{ and } n(1 - \hat{p}) \geq 10$$

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

All n or p

$$\tilde{p} \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}}$$

100(1- α) CIs around the difference between two population proportions

X: Number of successes in n trials, each trial is i.i.d. with constant $p=P\{\text{success}\}$

We are looking for a random variable , $Y=h(X; \theta)$ whose distribution can be specified.

$$X \geq 10, n - X \geq 10, Y \geq 10, m - Y \geq 10$$

$$\hat{p}_X - \hat{p}_Y \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_X(1 - \hat{p}_X)}{n} + \frac{\hat{p}_Y(1 - \hat{p}_Y)}{m}}$$

All n,m, X, Y

$$\tilde{p}_X - \tilde{p}_Y \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}_X(1 - \tilde{p}_X)}{\tilde{n}} + \frac{\tilde{p}_Y(1 - \tilde{p}_Y)}{\tilde{m}}}$$

$$\hat{P}_X = \frac{X}{n}, \hat{P}_Y = \frac{Y}{m}$$

$$\tilde{P}_X = \frac{X+1}{\tilde{n}}, \tilde{n} = n + 2, \tilde{P}_Y = \frac{Y+1}{\tilde{m}}, \tilde{m} = m + 2$$

Prediction interval (PI) for a single future value

$X_1, \dots, X_n : N(\mu, \sigma)$ random sample with unknown μ and σ .

Goal: to predict a single value of a future X , which is a **random variable**

In CI, goal: to estimate the mean value of that random variable, which is unknown but **constant**

Point predictor for X_{n+1} : $\bar{X}$

Prediction error: $\bar{X} - X_{n+1} \sim \text{Normal}$

Expected value of prediction error: $E(\bar{X} - X_{n+1}) = \dots$

Variance of prediction error: $V(\bar{X} - X_{n+1}) = \dots$

Theorem

Let $\bar{X}$ and S be the mean and sample standard deviation, respectively, of a random sample of size n from a normal distribution with mean μ . Then, the rv

$$T_{n-1} = \frac{\bar{X} - X_{n+1}}{S\sqrt{1 + 1/n}}$$

has a t distribution with $n-1$ degrees of freedom (df).

Proposition

$$T_{n-1} = \frac{\bar{X} - X_{n+1}}{s\sqrt{1 + 1/n}}$$

Let $\bar{x}$ and s be the sample mean and sample standard deviation computed from the results of a random sample of n from a normal population with mean μ . Then, a prediction interval (PI) for a single observation to be selected from a normal population distribution with prediction level $100(1-\alpha)\%$ is:

$$\left(\bar{x} \pm t_{n-1, \alpha/2} s \sqrt{1 + \frac{1}{n}} \right)$$

CI & PI as $n \rightarrow \infty$

As $n \rightarrow \infty$:

$$\bar{X} \rightarrow \mu$$

$$S \rightarrow \sigma$$

$$CI: \left(\bar{x} \pm t_{n-1, \alpha/2} S \sqrt{\frac{1}{n}} \right)$$

$$PI: \left(\bar{x} \pm t_{n-1, \alpha/2} S \sqrt{1 + \frac{1}{n}} \right)$$

$$S \sqrt{1/n} \rightarrow 0 \quad \Rightarrow \quad CI \rightarrow \mu$$

$$S \sqrt{1 + 1/n} \rightarrow \sigma \quad \Rightarrow \quad PI \rightarrow (\mu \pm z_{\alpha/2} \sigma)$$

Tolerance Intervals for a Normal Population

A **tolerance interval** is an interval that is likely to contain a specified proportion of the population.

The method we will use requires that the population be normal.

Tolerance Intervals

Let $X_1, \dots, X_n$ be a sample from a *normal* population.

A tolerance interval for containing $100(1 - \gamma)\%$ of the population with confidence $100(1 - \alpha)\%$ is:

$$\bar{X} \pm k_{n,\alpha,\gamma} S$$



Table A.4

$$100(1 - \alpha)\%$$

$$100(1 - \gamma)\%$$

Table A.4

TABLE A.4 Tolerance factors for the normal distribution

Sample Size n	Confidence Level 95%			Confidence Level 99%		
	Percent of Population Contained			Percent of Population Contained		
	90%	95%	99%	90%	95%	99%
2	32.0187	37.6746	48.4296	160.1940	188.4915	242.3004
3	8.3795	9.9158	12.8613	18.9304	22.4009	29.0553
4	5.3692	6.3699	8.2993	9.3984	11.1501	14.5274
5	4.2749	5.0787	6.6338	6.6118	7.8550	10.2602
6	3.7123	4.4140	5.7746	5.3366	6.3453	8.3013
7	3.3686	4.0074	5.2481	4.6129	5.4877	7.1868
8	3.1358	3.7317	4.8907	4.1473	4.9355	6.4683
9	2.9670	3.5317	4.6310	3.8223	4.5499	5.9660
10	2.8385	3.3794	4.4330	3.5821	4.2647	5.5943
11	2.7372	3.2592	4.2766	3.3970	4.0449	5.3075
12	2.6550	3.1617	4.1496	3.2497	3.8700	5.0792
13	2.5868	3.0808	4.0441	3.1295	3.7271	4.8926
14	2.5292	3.0124	3.9549	3.0294	3.6081	4.7371
15	2.4799	2.9538	3.8785	2.9446	3.5073	4.6053
16	2.4371	2.9029	3.8121	2.8717	3.4207	4.4920
17	2.3995	2.8583	3.7538	2.8084	3.3453	4.3934
18	2.3662	2.8188	3.7022	2.7527	3.2792	4.3068
19	2.3366	2.7835	3.6560	2.7034	3.2205	4.2300
20	2.3099	2.7518	3.6146	2.6594	3.1681	4.1614
25	2.2083	2.6310	3.4565	2.4941	2.9715	3.9039
30	2.1398	2.5494	3.3497	2.3848	2.8414	3.7333
35	2.0899	2.4900	3.2719	2.3063	2.7479	3.6107
40	2.0516	2.4445	3.2122	2.2468	2.6770	3.5177
45	2.0212	2.4083	3.1647	2.1998	2.6211	3.4443
50	1.9964	2.3787	3.1259	2.1616	2.5756	3.3846
60	1.9578	2.3328	3.0657	2.1029	2.5057	3.2929
70	1.9291	2.2987	3.0208	2.0596	2.4541	3.2251
80	1.9068	2.2720	2.9859	2.0260	2.4141	3.1725
90	1.8887	2.2506	2.9577	1.9990	2.3819	3.1303
100	1.8738	2.2328	2.9343	1.9768	2.3555	3.0955
200	1.7981	2.1425	2.8158	1.8651	2.2224	2.9207
300	1.7670	2.1055	2.7671	1.8199	2.1685	2.8499
400	1.7492	2.0843	2.7392	1.7940	2.1377	2.8094
500	1.7373	2.0701	2.7206	1.7769	2.1173	2.7826
600	1.7287	2.0598	2.7071	1.7644	2.1024	2.7631
700	1.7220	2.0519	2.6967	1.7549	2.0911	2.7481



Example

The lengths of bolts manufactured by a certain process are known to be normally distributed. In a sample of 30 bolts, the average length was 10.25 cm, with a standard deviation of 0.20 cm. Find a tolerance interval that includes 90% of the lengths of the bolts with 95% confidence.

$$\bar{X} \pm k_{n,\alpha,\gamma} S$$

$$10.25 \pm 2.1398 \times 0.2$$

$$n=30$$

$$100(1 - \gamma)\% = 90\% \rightarrow \gamma = 10\%$$

$$100(1 - \alpha)\% = 95\% \rightarrow \alpha = 5\%$$

$$(9.82, 10.68)$$



$$100(1 - \alpha)\% = 95\%$$

$$100(1 - \gamma)\% = 90\%$$

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4	5.3692	6.3699	8.2993	9.3984	11.1501	14.5274
5	4.2749	5.0787	6.6338	6.6118	7.8550	10.2602
6	3.7123	4.4140	5.7746	5.3366	6.3453	8.3013
7	3.3686	4.0074	5.2481	4.6129	5.4877	7.1868
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25	2.2083	2.6310	3.4565	2.4941	2.9715	3.9039
30	2.1398	2.5494	3.3497	2.3848	2.8414	3.7333
35	2.0899	2.4900	3.2719	2.3063	2.7479	3.6107
40	2.0516	2.4445	3.2122	2.2468	2.6770	3.5177
45	2.0212	2.4083	3.1647	2.1998	2.6211	3.4443
50	1.9964	2.3787	3.1259	2.1616	2.5756	3.3846
60	1.9578	2.3328	3.0657	2.1029	2.5057	3.2929
70	1.9291	2.2987	3.0208	2.0596	2.4541	3.2251
80	1.9068	2.2720	2.9859	2.0260	2.4141	3.1725
90	1.8887	2.2506	2.9577	1.9990	2.3819	3.1303
100	1.8738	2.2328	2.9343	1.9768	2.3555	3.0955
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300	1.7670	2.1055	2.7671	1.8199	2.1685	2.8499
400	1.7492	2.0843	2.7392	1.7940	2.1377	2.8094
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600	1.7287	2.0598	2.7071	1.7644	2.1024	2.7631
700	1.7220	2.0519	2.6967	1.7549	2.0911	2.7481

Python for Tolerance Intervals

```
def tolerance_intervals(column, percentile, conf_level, sigma):  
  
    z_score = norm.ppf(percentile)  
  
    t = nct.ppf(conf_level, df=len(column)-1., nc=np.sqrt(len(column))*z_score)  
  
    sample_mean=np.mean(column)  
  
    k = t / np.sqrt(len(column))  
  
    conf_lower_bound = sample_mean - (k*sigma)  
  
    return conf_lower_bound
```

Class exercise

A manufacturer produces a certain type of battery. The quality standards require the capacity of these batteries to be in between (137, 143) ampere-hours.

To meet these standards, the production process managers decided to design the production process such that:

- Let μ be the population average of the battery capacities. The production process should ensure that $\mu = 140$.
- Let σ^2 be the population variance of the battery capacities. The production process should ensure that $\sigma^2 = 4$.
- Let p be the proportion of batteries that meet the quality standards. The production process should ensure that $p = 0.95$.

Class exercise

An independent consumer-protection agency wishes to test whether the manufacturer complies to the quality standards. For this purpose, they measure the capacity of a random sample of 20 batteries from a recently produced batch. They compiled the following information from this sample.

Note that the normal random sample consists of $(X_1, X_2, \dots, X_{20})$:

- X_i denotes the capacity of i^{th} battery.
- Y denotes the number of batteries complying the quality standards.
- $n=20$ is the sample size.

Class exercise

Parameter	95% CI	95% PI	95% TI for 90%
μ	(137.226, 139.714)	(135.16, 141.78)	(123.32, 153.62)
σ^2	(4.09, 15.08)	---	---
p	(0.43, 0.82)	---	---

1. Does the manufacturer meet the quality standards? Why or why not? Explain.
2. Does the production process satisfy the managers' design? Why or why not? Explain.
3. If the production process satisfied the managers' design, do you think it would also meet the quality standards? Why or why not? Explain.

- Capacity of these batteries to be in (137, 143) ampere-hours
- Let μ be the population average of the battery capacities. The production process should ensure that $\mu = 140$.
- Let σ^2 be the population variance of the battery capacities. The production process should ensure that $\sigma^2 = 4$.
- Let p be the proportion of batteries that meet the quality standards. The production process should ensure that $p = 0.95$.

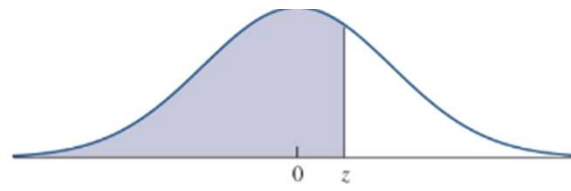
Class exercise

$(X_1, X_2, \dots, X_{20})$: normal random sample

If the design works:

- $\mu = 140, \sigma^2 = 4$. Can we have $p = 0.95$?
- $\mu = 140, p = 0.95$. What should the design aim for σ^2 ?





z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

